Project Ditto v4 — Pre-Registration Specification

Specification v1.0 · Pre-registration draft Full PDF: `SPEC.pdf` (immutable — to be generated and committed after author review)

> **This document will be frozen before any chain construction begins.** Thresholds, methodology, and analysis plan are committed here before implementation. Any methodology change after pre-registration commit invalidates the pre-registration and requires a new dated spec. Pre-registration discipline is matched to v3's level (`SPEC.md` + immutable `SPEC.pdf` + dated supplements for any post-commit amendments).

Hypothesis (precisely scoped)

When v3's pre-registered methodology — the prompt template `v3.1-game` , the state-signature design from `src/reference.py` , the both-actionable filter logic, and the paired-McNemar / paired-t / Bonferroni statistical analysis — is applied to the constraint chains constructed by v1 (Project-Ditto Pokémon Showdown telemetry), with a single v1-domain adaptation to the actionable-types set (see §"Reference Distribution Build"), the evaluation will reproduce v1's published moderate-positive Haiku result.

The directional prediction is:

> **Layer 1 actionable gap on Haiku 4.5 will clear the moderate-positive threshold (gap ≥ 0.05, Bonferroni-corrected $p < 0.05$) on v1's chain data evaluated under v3's methodology.**

This is a single-cell methodology characterization study. It is not a replication of v1, a re-analysis of v3, a transfer experiment, or a test of the original training-exposure hypothesis. The question it isolates is narrow and specific:

> *Does v3's pre-registered methodology produce signal on chains generated under v1's organically-distributed construction process?*

If yes (Haiku Layer 1 actionable gap clears moderate-positive on v1 chains under v3 methodology): v3's pre-registered methodology is sound; v3's reversed result on three of its four formal-game cells is most likely attributable to v3-specific chain-construction properties rather than to the methodology itself.

If no (the gap fails to clear, or reverses): v3's pre-registered methodology has issues that affect even chains v1 originally showed strong signal on. The methodology requires more substantive review than the audit-pattern fixes already applied via `SPEC_v1.1.md` .

This is the only question v4 answers. v4 does not test post-hoc corrections to v3's methodology, does not characterize the four-outcome joint behavior of pre-registered and corrected methodologies, and does not produce statistical evidence supporting any specific corrected- methodology design. Those questions remain open after v4 and require separate experiments to address.

What this experiment is NOT

This list is exhaustive on what v4 can and cannot conclude. The list is load-bearing for interpretation; deviations from it after pre- registration commit are spec amendments.

- **NOT a replication of v1.** v1's published numbers stand independently on v1's pre-registered methodology. v4 does not validate v1's publication; v1 is treated as a fixed reference point with a known outcome (Haiku gap 0.066 corrected, Sonnet gap 0.206 corrected, per `CORRECTED_SCORING.md` in the v1 repo).
- **NOT a re-analysis of v3.** v3's pre-registered classification (`reversed` on chess_standard, chess960, draughts_intl; outcome on checkers_american is methodology-sensitive — see §"v3 status as of v4 pre-registration" below) is unchanged regardless of v4's outcome. v4 does not have the standing to retroactively reclassify v3.
- **NOT a transfer test.** No claim is made about whether v3's models trained on v3's data, whether v1's models trained on v1's data, or about cross-domain generalization of any abstraction.
- **NOT a test of the training-exposure hypothesis.** v3's hypothesis was that training-data exposure compresses the real-vs-shuffled detectability gap (with chess_standard / chess960 and checkers_american / draughts_intl as within-family contrasts). v4 does not test this hypothesis. v1's domain has no within-family exposure contrast.
- **NOT a validation of any post-hoc methodology corrections.** v4 runs v3's pre-registered methodology only. No comparison to corrected variants is performed. Any claim about whether post-hoc corrections to v3's methodology recover signal is out of scope; it requires a separate experiment with a corrected methodology specified in advance.
- **NOT additive to v1's findings.** v4 produces no new claim about Pokémon, about v1's models, or about v1's prompt format. v4's contribution is purely diagnostic: characterizing v3's methodology on chains of known structure.
- **NOT a multi-cell experiment.** A single source (v1 Pokémon chains) is evaluated. There is no within-family contrast and no Bonferroni family across cells.

Why we are running this experiment now

This section describes the program context honestly so that a future reader can understand v4's place in the program.

v1 (Project-Ditto, Pokémon Showdown telemetry; published 2026-04-21, methodology correction 2026-04-25): pre-registered methodology produced strong-positive on Sonnet (gap 0.206 corrected) and moderate-positive on

Haiku (gap 0.066 corrected). Both findings cleared the corrected paired- test methodology with extreme statistical margins.

v2 (Project-Ditto-v2, programming-task agent trajectories; published 2026-04-25): pre-registered methodology partially replicated v1's effect. Layer 2 cleared on all four model × source cells; Layer 1 actionable on Haiku-TB cleared the moderate-positive threshold (gap 0.0801, Bonferroni- corrected p = 0.0116). The other three cells did not clear. Direction positive in 7 of 8 cells.

v3 (Project-Ditto-V3, formal-rule games; pre-registered 2026-04-25, Phase 1 evaluated 2026-04-25, scored shortly after): under v3's pre- registered methodology with the SPEC_v1.1 amendments applied (entity- only prompt v3.1-game, focal_action cutoff fix, scorer output structure separation), Phase 1 Haiku produced negative Layer 1 actionable gaps on three of four cells (chess_standard −0.187, chess960 −0.231, draughts_intl −0.155). The fourth cell, checkers_american, returned a positive unfiltered gap (+0.039) that flipped to negative under the both-actionable filter (−0.115); see v3 SESSION_LOG.md "v4 Cell 1 — Statistical methodology robustness" for the full per-methodology table.

A v4 Cell 1 analysis on a non-merged branch in v3's repository characterized the robustness of v3's outcome to four statistical methodology choices; that work is referenced in §"Relationship to v4 Cell 1" below. Diagnostic mechanisms identified during v3 post-hoc analysis (resource_side dominance and backoff differential are recorded in v3's SESSION_LOG; further mechanism analysis if any is not yet documented in materials available at v4 pre-registration time) trace to v3-specific design choices in the formal-game T-code.

The question v4 isolates is whether v3's *methodology* — independent of v3's *chains* — produces the predicted detectability gap on chains already known to carry the relevant structure under a different pre-registered methodology. If methodology-on-clean-chains succeeds, the program's inferential picture stabilizes around chain-construction as the source of v3's reversal. If methodology-on-clean-chains fails, the methodology requires more substantive review.

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v3 status as of v4 pre-registration

This subsection records v3's status at the time of v4 pre-registration to anchor v4's interpretation framework. Numbers are sourced from v3's SESSION_LOG.md "v4 Cell 1 — Statistical methodology robustness" entry (2026-04-26).

v3 cell	Pre-registered methodology gap	v3 classification
chess_standard	−0.187	reversed (robust across 4 methodologies)
chess960	−0.231	reversed (robust across 4 methodologies)
checkers_american	−0.115 (filtered) / +0.039 (unfiltered)	methodology-dependent; under review

v3 cell	Pre-registered methodology gap	v3 classification
draughts_intl	−0.155	reversed (filter+Bonferroni-dependent significance)

v3's pre-registered classification per its SPEC.md is reversed on three of four cells under the pre-registered methodology, with checkers_american flagged for co-author review. v4's analysis does not depend on the disposition of checkers_american; v4 references v3 only as the methodology source under test.

Pre-registered Success Criteria

Per-cell thresholds (single cell: Haiku × v1 Pokémon chains)

Criterion	Threshold	Tier
Layer 1 actionable gap (real – shuffled, both-actionable filter)	≥ 0.05	Moderate-positive
Layer 1 actionable significance	Bonferroni-corrected p < 0.05	Moderate-positive
Layer 2 gap (legality × optimality composite)	≥ 0.04	Layer 2 confirmation
Layer 2 significance	Bonferroni-corrected p < 0.05	Layer 2 confirmation
Strong-positive	Layer 1 actionable gap ≥ 0.08 ∧ Bonferroni p < 0.01	Strong-positive

These thresholds match v3's SPEC.md exactly. The Bonferroni divisor for v4 is **1** — a single analysis (v3's pre-registered methodology) on a single cell (Haiku × v1 Pokémon chains). Bonferroni correction with divisor 1 is equivalent to no correction; the column is retained in the results output for cross-experiment comparability with v1/v2/v3 reporting conventions but applies no actual correction. The "Bonferroni-corrected p" reported in §"Pre-registered Success Criteria" is therefore identical to the raw McNemar p in v4's output.

Outcome tiers (consistent with v1, v2, v3): `strong_positive`, `moderate_positive`, `weak_mixed`, `null`, `reversed`.

Threshold for "v3 methodology works on clean chains"

The v4 hypothesis is supported iff the **primary** analysis (v3's pre- registered methodology) clears at least the moderate-positive threshold. Strong-positive on the primary is a stronger form of support but is not required for v4 to be informative.

A null result on the primary (gap not clearing 0.05 or p not clearing Bonferroni 0.05) refutes the v4 hypothesis. A reversed result on the primary (gap negative + Bonferroni-significant) is the strongest form of refutation and would imply that v3's methodology produces inversion even on chains v1 originally showed strong signal on.

Pre-registered Interpretation Framework

The four outcomes below are committed before any data collection. They correspond to the four possible classifications of the primary analysis result. v4 runs a single primary analysis (v3's pre-registered methodology) on a single cell (Haiku × v1 Pokémon chains). There is no secondary analysis and no joint interpretation across analyses.

Four-outcome interpretation

Primary: strong-positive (Layer 1 actionable gap ≥ 0.08, $p < 0.01$). v3's pre-registered methodology — applied to v1's chains under the two pre-registered v1-domain adaptations — produces a stronger detectability gap than v1's own pre-registered methodology produced on Haiku (v1 published Haiku gap 0.066 corrected). Strongest possible support for the v4 hypothesis: the methodology is not the source of v3's reversed result. v3's reversal is most parsimoniously explained by v3-specific chain-construction properties.

Primary: moderate-positive (Layer 1 actionable gap ≥ 0.05, $p < 0.05$). v3's pre-registered methodology produces a detectability gap on v1 chains in the same tier as v1's own published Haiku result. Supports the v4 hypothesis. v3's reversal is plausibly attributable to chain construction rather than methodology.

Primary: null (gap not clearing 0.05, or p not clearing 0.05; gap direction not specifically negative). v3's pre-registered methodology fails to detect signal on chains v1 originally classified as moderate-positive. Refutes the v4 hypothesis. The methodology has issues affecting chains beyond v3's domain. Possible explanations: (a) the SPEC_v1.1 amendments to v3 (entity-only prompt, focal_action fix, scorer output structure) are insufficient to rehabilitate the methodology; (b) v1's signal is specific to v1's prompt/scorer combination and does not survive the v3.1-game prompt reformulation even with chain content held constant. v4 alone cannot distinguish (a) from (b); a follow-up experiment with v3 methodology + v1 prompt, or v1 methodology + v3.1 prompt, would be required to disambiguate.

Primary: reversed (gap negative, Bonferroni-corrected $p < 0.05$ in the negative direction). Strongest possible refutation of the v4 hypothesis. v3's pre-registered methodology produces *inverted* signal on chains v1 classified as moderate-positive. The methodology has structural issues that affect even chains of known good structure. A substantial review of v3's scoring pipeline is required before any follow-on experiment.

What v4 does NOT change

v3's pre-registered classification stands. Regardless of v4's outcome, v3's pre-registered classification per its own SPEC.md is unchanged. v4 has standing to comment on v3's *methodology*, not on v3's *classification* — those are different objects.

v1's published numbers stand. v4 uses v1 chains as input but does not re-evaluate v1's published gaps. v1's results are a fixed reference point; v4's outcome carries no implication for v1's standing.

Models and Evaluation Parameters

Parameter	Value
Models	Claude Haiku 4.5 (`claude-haiku-4-5-20251001`) — only model
Primary config	Temperature 0.0, seed 42
Variance study	T=0.5, seed 1337; T=0.5, seed 7919
Max tokens	50
Cutoff K	`len(constraints) // 2` (matches v1 and v3 — `max(1, …)` floor applied per `reference.py` defensive logic)
Prompt version	v3.1-game (verbatim from v3 `src/prompt_builder.py` ; system + user prompts unchanged)
API	Anthropic Messages Batches (50% cost reduction)
Action normalization	Full normalization (case, punctuation, whitespace, word order) — same as v1/v2/v3

Model selection rationale (Haiku-only)

v4 is a single-cell methodology characterization study answering a binary question. v3's Phase 1 ran Haiku to gate the more expensive Sonnet phase. v4 mirrors v3's Phase 1 structure. Sonnet is not run because:

- The binary "does v3's methodology produce signal on clean chains?" question is fully answerable with Haiku.
- Adding Sonnet doubles cost and runtime without changing what v4 can conclude.
- v1 published Haiku gap 0.066 (moderate-positive). v4 tests whether v3 methodology reproduces *that* number's tier, not the larger Sonnet effect. The Haiku comparison is the directly relevant one.

If v4 primary returns null or reversed, a follow-up Sonnet evaluation may be warranted — but that is a separate experiment, not part of v4's pre-registration.

Prompt portability check (verbatim use)

v3.1-game prompt body:

```
 Given the state above, what is the most likely entity (resource, tool,
phase, or coordination tag) at step {cutoff_k + 1}?
Output only a single token like piece_A, chain_B, formation_C,
phase_endgame, or progress_remaining.
No verbs, no explanation.
```

System prompt: v3's `SYSTEM_PROMPT` from `src/prompt_builder.py` (framing: "sequential decision process", domain-blind throughout).

The prompt examples reference `piece_A`, `chain_B`, `formation_C`, `phase_endgame`, `progress_remaining` — these are v3-domain entity labels. They are retained **unchanged** in v4. Rationale: v4's hypothesis is that v3's *methodology unchanged* produces signal on clean chains. Modifying the prompt examples for Pokémon would change the methodology under test. The model receives v3-flavored entity example labels paired with a chain prefix containing v1-flavored entity labels (`unit_A`, `action_3`, etc.); whether the model's output distribution adjusts to the prefix's vocabulary is part of what v4 is implicitly testing. If pilot inspection (§"Phase structure") shows near-zero match rate driven by example-vocabulary fixation rather than a real methodology issue, this is a pre-pilot risk to flag, not a post-pilot spec amendment.

The `sequential decision process` framing in the system prompt is fully domain-blind and applies without modification.

Data Sources (committed — will not change after freeze)

Primary source: v1 Pokémon chains

Property	Value
Source	`chains/real/` directory of v1 repo (Project-Ditto, Pokémon Showdown telemetry)
Chain construction	v1 pipeline frozen at git tag `T-v1.1-frozen` (per v1 CLAUDE.md)
Chain schema	v1 JSONL: per-file chain dict with `constraints` (list of constraint dicts), `cutoff_k`, `chain_id`, etc. (per v1 CLAUDE.md and confirmed against v1 reference.py interface)
Per-cell sample target	1,200 real chains
Shuffled variants	3 per real chain at seeds 42, 1337, 7919 (v1 default; matches v3's shuffler)

Sample size: 1,200 v1 Pokémon chains

Matches v3's per-cell sample target. Power calculations from v3 SPEC.md §"Sample Size & Power" carry over and are **stronger** at v4's smaller Bonferroni divisor: at $\alpha = 0.05$ (v4 divisor = 1) with $n = 1{,}000$ paired evaluations and v2-calibrated discordant rates, per-cell power for moderate-positive (gap ≥ 0.05) exceeds the ~94% v3 calculated at $\alpha = 0.0125$ (v3 Phase 1 divisor = 4). Power for strong-positive remains $\geq 99\%$. v4 is generously powered for the binary hypothesis.

The 1,200-chain target assumes v1's chains achieve approximately the same both-actionable-retention rate v3 calibrated against (~83% retention = ~1,000 paired evaluations after filter). v1's chain set is large enough that

1,200 chains can be sampled without re-running v1's acquisition pipeline (v1 reports having generated 4,806 real chains across p1+p2 perspectives per `CORRECTED_SCORING.md` data structure).

Sampling is deterministic at seed 42 from v1's available chains, with the seed and the ordered list of selected `chain_id`s recorded in `v4_chain_selection.json` and committed before any evaluation runs.

What is NOT used

- v1's reference distribution (`reference_dist.pkl`) is **not used**. v4 builds a fresh reference distribution from v1's chains using v3's `src/reference.py` (with the v1-domain adaptations described in §"Reference Distribution Build"). Using v1's own reference would mean evaluating v1 chains under a hybrid methodology, defeating v4's test-v3-methodology framing.
- v1's prompt template, scorer, runner, T-code are not used. v4 uses v3's versions throughout.
- v2's modules and chains are not used.

Reference Distribution Build

This section implements the original prompt's flagged "biggest risk" explicitly. The ground truth is that v1's StateSignature (`(active_pair, hp_brackets, status_effects, field_conditions, turn_bucket)`) and v3's StateSignature (`(current_phase, last_move_type, resource_bracket, entity_label)` at level 0) have different shapes. v4 must choose between using v1's or v3's signature design. The pre-committed choice is:

> **v4 uses v3's StateSignature design with the two v1-domain adaptations documented below (phase-name default; actionable types include InformationState), building a fresh reference distribution from v1 chains.** The signature *structure* (4-level backoff, `(current_phase, last_move_type, resource_bracket, entity_label)`, `extract_entity_from_constraint` logic) is unchanged from v3.

v3's `src/reference.py` is largely domain-blind

Inspection of v3's `src/reference.py` confirms that its components are mostly domain-blind by design:

Component	Source	v1 chain compatibility
`current_phase`	last `SubGoalTransition.to_phase` in prefix; defaults to `"phase_opening"`	v1 chains have `SubGoalTransition` constraints. Default may need adaptation — see below.
`last_move_type`	`constraint_type` of `constraints[cutoff_k - 1]` (last shown)	Domain-blind: any constraint type name. v1's chains

Component	Source	v1 chain compatibility
		use the same six types.
`resource_bracket`	bracket of last `ResourceBudget.amount` (0–4 buckets)	Domain-blind: `_resource_bracket` operates on a normalized float. v1 chains have `ResourceBudget` constraints with `amount` fields.
`entity_label`	`extract_entity_from_constraint(constraints[cutoff_k])`	Domain-blind: returns whatever lowercased abstract label the constraint dict carries (e.g. v3's `piece_g`, v1's `unit_a`).

`extract_entity_from_constraint` itself is fully domain-blind; it reads known fields off six constraint types and lowercases the result. v1's constraints carry the same fields (`tool` , `to_phase` , `resource` , `dependency` , `objective` , `observable_added`).

Required adaptation: phase-name default

v3's `current_phase` defaults to the string `"phase_opening"` when no `SubGoalTransition` appears in the prefix window. This default is v3-specific (chess/checkers phase taxonomy: opening / middlegame / endgame). v1's `to_phase` taxonomy is different (Pokémon battle phases emitted by v1's T-code).

Adaptation: v4 retains v3's `extract_state_signature` function unchanged in structure, but the **default phase string** is selected empirically from the pilot 50-chain set. Pre-registration commits to: *the default value maximizing level-0/1 reference coverage on the pilot set, recorded in* `v4_chain_selection.json` *before the full evaluation runs*. If multiple candidate defaults tie within 0.5 percentage points of coverage, the lead author selects (and records the selection in the pilot session log). The default value is a *parameter*, not a methodology change — v3's `reference.py` itself documents this default as `# Defensive default`.

Required adaptation: actionable types include InformationState

v3's both-actionable filter excludes `InformationState` because formal games are perfect-information. v1's domain (Pokémon battles) has meaningful `InformationState` — opponent Pokémon are hidden until revealed; opponent HP is bucketed; opponent moves are hidden until used. Excluding `InformationState` from v4's actionable set would remove a substantial fraction of v1's actionable chain content and materially distort the test.

Pre-committed adaptation: v4's `ACTIONABLE_TYPES` set is:

```
{
  "ToolAvailability",
  "SubGoalTransition",
  "InformationState",   # added (v1-domain — meaningful hidden info)
  "CoordinationDependency",
  "OptimizationCriterion",
}
```

This is **five** actionable types. v3 uses four (excludes both `InformationState` and `ResourceBudget`). v1's published methodology did not apply an actionable-types filter at all (per v1 `CORRECTED_SCORING.md` : "v1's filter rule (reference-miss exclusion) is preserved... v2's actionable-constraint-type filter is **not** imported; v1 had no such filter"). v4's five-type set is the v3 filter logic with `InformationState` re-included for v1's domain. `ResourceBudget` remains excluded as non-actionable, matching v2's and v3's design choice for material-counting / budget constraints that the model cannot meaningfully predict in advance.

This is a v1-domain adaptation, **not** a methodology change in v3's sense. The pre-registered scorer logic operates on whatever set is configured for `ACTIONABLE_TYPES` ; v4 merely configures the set appropriate to v1's domain. The adaptation is documented here to be explicit; the SPEC commits to it.

Coverage target

v3 SPEC.md §"Pipeline Reuse from v1/v2" specifies ≥ 90% non-max-backoff coverage. v4 retains this target. Pilot validation (§"Phase structure") verifies coverage before the full evaluation runs.

Pre-registered Statistical Methodology

Primary (and only) analysis: v3's pre-registered methodology with v1-domain adaptations

Layer 1 (primary): paired McNemar's test with continuity correction, both-actionable filter (with `ACTIONABLE_TYPES` set per §"Reference Distribution Build" above). Bonferroni divisor = 1 (single analysis; no correction applied). The threshold is therefore raw McNemar $p < 0.05$ for moderate-positive and raw McNemar $p < 0.01$ for strong-positive.

Layer 2: paired t-test (`scipy.stats.ttest_rel`) on legality × optimality composite scores. Same Bonferroni treatment (divisor = 1).

Pair alignment by `(base_chain_id, eval_seed)` — model dimension is fixed (Haiku only). Three shuffled variants per real chain (seeds 42, 1337, 7919) produce three pairs per real evaluation, matching v3's `(base_chain_id, model, eval_seed) × (shuffled_chain = base_chain_id + "_shuffled_" + shuffle_seed)` pair structure.

Pairs missing either real or shuffled result are reported and excluded. Gap is computed as `real_match_rate - shuffled_match_rate` over all included pairs (matching v2/v3 estimand convention).

The scorer is v3's `src/scorer.py` (per `SPEC_v1.1.md` Amendment 3, producing the `primary_cells` / `variance_study` separation), with two configuration changes:

1. `SOURCES = ["pokemon"]` (single cell)
2. `ACTIONABLE_TYPES` includes `InformationState` (5-type set per §"Reference Distribution Build")

The scorer's statistical logic is unchanged. The per-cell threshold classification logic is unchanged. The output schema's `primary_cells` key contains a single entry (`haiku::pokemon`); the `variance_study` key contains two entries (`haiku::pokemon::T0.5_seed1337` , `haiku::pokemon::T0.5_seed7919`).

No secondary analysis

v4 does not include a comparison to a post-hoc-corrected methodology. This is a deliberate scope decision recorded at pre-registration:

- The corrected methodology referenced in v4's planning prompt ("downweighted reference + matched-backoff stratification + leave- one-out, as developed in v3's Phase A and Path 1 analyses") is not implementable from documents available at v4 pre-registration time.
- Including a secondary analysis whose specification is incomplete would not be a legitimate pre-registration.
- Even if the secondary methodology were specifiable, v4 could not *validate* it (the corrections were designed against v3 data; v1 data observing them work would be consistent with the diagnosis but would not independently establish the corrections' validity).

A separate experiment with a fully-specified corrected methodology, pre-registered against data that was not used in the methodology's design, would be required to produce evidence about post-hoc corrections. v4 does not attempt that.

Phase Structure

v4 has no formal gates between phases (the experiment is small and the question is binary). The phase structure below is a sequencing aid, not a gating mechanism.

Phase 0 — Pre-registration commit

- Author review of this SPEC.md (lead + co-author)
- Verification of v1 chain mirror in `data/v1_chains_mirror/` and commit of the deterministic 1,200-chain selection at seed 42 (recorded in `v4_chain_selection.json`)
- Verification of v3 git submodule pinned to `T-code-game-v1.0-frozen`
- Generation and commit of `SPEC.pdf` from finalized `SPEC.md`
- Pre-registration commit message: "Pre-registration commit — thresholds and methodology frozen"
- Both-author sign-off recorded in commit history (matching v3's protocol)

Phase 1 — Pilot validation (50 chains)

50 v1 chains evaluated end-to-end under the v3 methodology. Pilot verifies:

- v1 chain JSONLs load correctly under v3's chain-handling code (compatibility check)
- Reference distribution build on v1 chains achieves ≥ 90% non-max- backoff coverage (matching v3's target); flag if below
- v3.1-game prompt produces parseable single-token responses on v1 chains at non-zero match rate (sanity check; not an effect-size measurement)
- Both-actionable filter retention rate is in the expected range (~60–85%); flag if outside

Pilot pass criteria: load OK; coverage ≥ 90%; non-zero match rate; both- actionable retention ≥ 50%.

Pilot fail response: pause for author review. Likely failure modes: phase-name default mismatch (adjust per §"Reference Distribution Build"), prompt-vocabulary fixation (flagged as a limit on v4's interpretation, not a spec amendment), chain-format incompatibility (technical fix).

Phase 2 — Full evaluation

1,200 v1 chains × (1 real + 3 shuffled variants) × 3 evaluation configs (T=0.0 seed=42; T=0.5 seed=1337; T=0.5 seed=7919) = **14,400 calls**. This matches v3's per-cell call volume (v3 ran 57,600 calls across 4 cells = 14,400 per cell).

Anthropic Messages Batches API. Cost estimate: $5–10 (Haiku batch rate of $0.40/M input + $2.00/M output; per v3 Session 10's actual cost of ~$17 on 57,600 calls scaled to v4's 14,400). The original planning prompt's $20–30 estimate was conservative; v4 should come in well under it.

Phase 3 — Scoring

Scorer is v3's `src/scorer.py` (per SPEC_v1.1 Amendment 3 output structure) with `SOURCES = ["pokemon"]` and the v1-domain `ACTIONABLE_TYPES` set. Scoring runs in a separate session from evaluation, on blinded inputs (model identity, condition, seed stripped before scoring), matching v3's blinded-scorer protocol.

Output: `results/v4_scored.json` containing per-config per-condition gap, McNemar χ^2, raw p, Bonferroni-corrected p, both-actionable retention rate, n_pairs, and outcome tier classification per the pre-registered thresholds.

Phase 4 — Write-up

Single results document `RESULTS.md` matching v3's RESULTS structure: abstract; pre-registration link; threshold check table; primary + secondary outcome cells with interpretation per the four-outcome framework; supplementary analyses (per-config variance; carrier analysis); discussion of limits; explicit statement that v4 does not change v3's pre-registered classification.

Co-author review before commit (matches v3 Gate 8 protocol).

Pre-registered Supplementary Analyses

1. **Per-config variance study** — three temperature/seed configs as above, reported as supplementary stability characterization. Not in the primary Bonferroni family, matching v3 SPEC.md §"Pre-registered Supplementary Analyses" item 1.

2. **Constraint-type carrier analysis** — which actionable constraint types carry the signal (or its inversion) on v1 chains under v3 methodology. Compares against v2's TB / SWE carrier asymmetry finding (`ToolAvailability` for TB, `InformationState` for SWE) and v1's per-type breakdown (which is in v1's results but was not the primary metric).

3. **Comparison to v1 published numbers** — v4 Haiku gap on v1 chains under v3 methodology, vs v1's published Haiku gap (0.066 corrected), vs v3's Phase 1 Haiku per-cell gaps (sourced from v3's `results/phase1_v31_scored_full.json` — same data v4 Cell 1's robustness analysis operated on, but used here only as descriptive reference points). Descriptive only; not a hypothesis test.

4. **Pair-level disagreement analysis** — for the small expected number of cases where v4 primary and v1's published methodology disagree at the chain level (real-correct under one and not the other), characterize the disagreement to support §"Constraint-type carrier analysis".

5. **Backoff-level distribution** — match rate per backoff level (0–3) for v1 chains under v4. Comparison to v3's per-cell backoff distributions (per v3's reference coverage report) is descriptive. This analysis is *deliberately not* a hypothesis test of the "backoff differential" mechanism; testing that mechanism on v4 data would require a Phase A-style diagnostic, which is out of scope for v4.

Relationship to v4 Cell 1 (statistical robustness analysis)

v3's SESSION_LOG.md contains an entry "v4 Cell 1 — Statistical methodology robustness — 2026-04-26" describing a study on a non- merged branch (`claude/v4-cell-1-statistical-robustness`) that applied four statistical methodologies to v3's existing primary-config response data. This SPEC describes the experiment that has been called **v4 Cell 2** in informal discussion: the v1-Pokémon-chains methodology characterization.

The two studies are independent:

- v4 Cell 1 holds v3's chains constant and varies the statistical methodology. It characterizes how robust v3's `reversed` outcome is to test-choice within v3's data.
- v4 Cell 2 (this SPEC) holds the methodology constant (v3's pre- registered methodology) and varies the chain construction (v3's formal-game chains → v1's Pokémon chains). It characterizes whether v3's methodology produces signal on chains generated under different, organically-distributed construction.

Both v4 cells inform the chain-vs-methodology question from different sides. Neither alone is sufficient; together they triangulate.

This SPEC does not commit to merging the v4 Cell 1 branch. That decision is independent of v4 Cell 2 and depends on co-author review of Cell 1's output (per v3 SESSION_LOG.md's standing recommendation).

What v4 would establish

If primary clears moderate-positive

> "v3's pre-registered methodology — applied to chains constructed under v1's organically-distributed pipeline, with the two pre-registered v1-domain adaptations (phase-name default; actionable types include `InformationState`) — produces a real-vs-shuffled detectability gap consistent with v1's own published Haiku result. This is consistent with the interpretation that v3's reversed result on three of four formal- game cells reflects v3-specific chain-construction properties (resource_side dominance and backoff differential per v3's SESSION_LOG; further mechanisms if any) rather than a structural problem with the v3 statistical methodology."

This does not validate v3's methodology in absolute terms; it only establishes that the methodology is not the proximate cause of v3's reversal.

If primary returns null or reversed

> "v3's pre-registered methodology fails to produce signal — or produces inverted signal — on chains v1's pre-registered methodology classified as moderate-positive. This implies the methodology has issues affecting chains beyond v3's domain, and the SPEC_v1.1 amendments (prompt vocabulary alignment, focal_action off-by-one, scorer output structure) are insufficient to fully rehabilitate it. v5 (or a v3 SPEC v1.2) requires a more substantial methodology review than the audit-pattern fixes already applied."

This does not invalidate v1's published result; v1 used a different methodology. v4's null/reversed result implies a methodology issue under v3's pre-registered design; it does not specify what would fix that issue. A follow-on experiment with v3 methodology + v1 prompt, or v1 methodology + v3.1 prompt, would help disambiguate whether the issue is in v3's prompt reformulation, in the focal_action / state-signature design, or elsewhere. v4 alone cannot make that determination.

Pre-registration discipline

Matching v3's discipline (per v3 CLAUDE.md):

- This `SPEC.md` is frozen at pre-registration commit.
- `SPEC.pdf` is the immutable anchor — not edited after the commit.
- Any post-commit amendment is recorded as a dated supplement (`SPEC_v1.1.md`, etc.) with both-author sign-off; `SPEC.md` is not overwritten.
- Pilot failure pauses execution for author review; auto-recovery is not permitted.
- Effect-size monitoring during evaluation runs is not permitted. Technical monitoring (API errors, malformed chains, missing pairs) is permitted.

Authors and Review

- Lead: Safiq Sindha (Microsoft Azure Hardware PM, independent research)
- Co-author: Myriam (Columbia University, systems engineering; Boeing FT) — reviews all major decisions; on all v4 publications

Pre-registration sign-off

- [x] **Safiq Sindha** (lead author) — date: 2026-04-26
- [x] **Myriam** (co-author, Columbia University) — date: 2026-04-26

Both authors have signed off on this SPEC v1.0. Pre-registration commit proceeds with this document and the generated SPEC.pdf as the immutable anchors. Any post-commit methodology change requires a dated supplement (`SPEC_v1.1.md`, `SPEC_v1.2.md`, etc.) with both-author sign-off; the SPEC.md and SPEC.pdf themselves are not modified after commit.

Pre-commit decisions (recorded; resolved by lead author 2026-04-26)

The following items were resolved before pre-registration commit:

1. **Secondary analysis: dropped** (Option 2.C). v4 runs primary analysis only. Bonferroni divisor = 1. v4 makes no claim about post-hoc-corrected methodologies.

2. **v1 chain availability: local mirror.** v1's chain JSONLs are available locally and will be mirrored into `data/v1_chains_mirror/` at pre-registration commit time. The deterministic 1,200-chain sample selected at seed 42 is recorded in `v4_chain_selection.json` and frozen at commit.

3. **Repository structure: git submodule.** v3 is referenced by git submodule pinned to its `T-code-game-v1.0-frozen` tag. v4 imports v3's modules from the submodule path; v3 source is not copied into v4. The submodule reference is part of the pre-registration commit.

4. **v4 Cell 1 relationship: parallel work, not input.** v4 Cell 2's RESULTS document references v4 Cell 1 as program context (existence, methodology-sensitivity finding) but does not condition v4 Cell 2's interpretation on Cell 1's disposition. Cell 2's outcome stands on its own pre-registration. Cell 1's eventual co-author review has no bearing on Cell 2's classification.

Empirical items pending pilot validation

The following item is empirical — pre-registration commits to the resolution rule, and the pilot data fills in the value.

1. **`current_phase` default value** (§"Reference Distribution Build"). v3's `extract_state_signature` defaults `current_phase` to `"phase_opening"` when no `SubGoalTransition` appears in the prefix window. v1's chains may use a different `to_phase` taxonomy. Pre-registration commits to: *"the default*

value maximizing level-0/1 reference coverage on the pilot 50-chain set, recorded in `v4_chain_selection.json` *before the full evaluation runs."* The pilot's reference-build step measures coverage at each candidate default; the highest-coverage value is selected. If multiple values tie within 0.5 percentage points, the lead author selects (and records the selection in the pilot session log).

Technical risks identified during drafting

The two HIGH-risk items at draft time (v1 chain accessibility, secondary analysis specification) were resolved by lead-author decisions recorded in §"Pre-commit decisions". The remaining risks are:

1. **Prompt-vocabulary fixation.** v3.1-game's prompt examples (`piece_A` , `chain_B` , `formation_C` , `phase_endgame` , `progress_remaining`) are formal-game-flavored. When evaluated on v1 Pokémon chains, the model may anchor on these example labels rather than picking up the prefix's vocabulary (`unit_A` , `action_3` , etc.), producing artifactually low match rates. Pilot validation surfaces this. If this materializes, v4 has a known interpretation limit: the "v3 methodology with two v1-domain adaptations" frame becomes "v3 methodology with two v1-domain adaptations *plus* v3-flavored prompt examples that may anchor the model away from v1-flavored prefix vocabulary." The SPEC's framing already accommodates this; an explicit note in RESULTS.md is warranted if the pattern is observed. **MEDIUM RISK; pilot will detect.**

2. **Phase-name default value.** v3's `current_phase` defaults to `"phase_opening"` . v1's chains may use different `to_phase` strings from `SubGoalTransition` constraints. If v1's `to_phase` taxonomy does not include any string that maps to a sensible v3-style default, the level-0/1 backoff hits drop dramatically and v4's reference coverage falls below the 90% target. Pilot detects. **MEDIUM RISK; mitigation in SPEC.**

3. **Both-actionable filter retention.** v4 adds `InformationState` to actionable types (v1-domain adaptation). This is a deliberate choice but it changes the filter's behavior relative to v3. If v1's `InformationState` constraints are themselves non-predictive (the model cannot meaningfully predict which Pokémon will be revealed next), including them in actionable could pull the gap toward zero. Pilot's actionable-retention measurement is one input; the carrier analysis (§Supplementary) is another. **LOW-MEDIUM RISK; documented as an interpretation note rather than a spec risk.**

4. **Stored-vs-recomputed focal_action.** v1 chains were generated under v1's chain-construction code, which may have stored `focal_action` per the original `cutoff_k - 1` interpretation (later fixed in v3 by SPEC_v1.1 Amendment 2 to be `constraints[cutoff_k]`). v3's `reference.py` now recomputes focal_action on-the-fly and ignores the stored field. v4's reference build uses v3's `reference.py` with the v1-domain adaptations (phase-name default, 5-type actionable set), so v4 inherits the on-the-fly focal_action computation. v1's stored `focal_action` is not read. This is correct but worth noting: if v1 chains' stored `focal_action` was used elsewhere in v1's published numbers, v4's gap will be computed against a slightly different focal-action target than v1's published gap. This is a feature, not a bug — v4 tests v3's methodology, which uses the corrected focal_action — but it means v4's gap is not directly comparable to v1's published gap on a numerical basis. The §"Comparison to v1 published numbers" supplementary analysis must call this out. **LOW RISK; documented.**

Estimated timeline

Per the §"Pre-commit decisions" resolutions, the only remaining gating item is pilot validation:

Phase	Calendar time	Notes
Pre-registration finalization	**DONE 2026-04-26**	Both authors signed off on SPEC v1.0
Pre-registration commit	< 1 day	Generate SPEC.pdf, initialize v4 repo (v3 submodule + v1 chain mirror + chain selection JSON), commit
Pilot (50 chains, end-to-end)	1–2 days	Includes reference build + scoring
Pilot review pause	1–3 days	Author review
Full evaluation (14,400 calls × 1 model)	< 1 day wall time	Anthropic Batches API; v3's 57,600-call run took 26 minutes, so v4's 14,400 calls should run in ~10–15 minutes
Scoring	< 1 day	Separate session, blinded
Write-up	3–5 days	RESULTS.md draft + co-author review
Total focused work	**1.5–2 weeks**	Matches the planning prompt's estimate

Budget: ~$5–10 in Haiku Batches API calls, well under the planning prompt's $20–30 estimate and well under prior phases.

Pre-registration v1.0 — drafted and signed off 2026-04-26 by Safiq Sindha (lead) and Myriam (co-author, Columbia University). Prior work: [Project Ditto v1] (Pokémon, moderate/strong positive), [Project Ditto v2] (programming, partial replication), [Project Ditto V3] (formal-rule games, reversed on three of four cells; checkers_american under review).